

Roshan Mishra

Full Stack Engineer — Bengaluru, India

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Full Stack Engineer with strong experience building scalable SaaS systems. Specialized in booking engines, rule-based availability systems, and high-performance REST APIs. Proven ability to translate complex business rules into reliable, production-grade software for product companies.

Professional Experience

- **Software Development Engineer (SDE-1), MONKSPACES.AI** (June 2023 - Present)
 - **Owned backend design and implementation of a production booking engine** handling the full rental lifecycle — lead intake, availability validation, unit allocation, payments, and contract creation — supporting nightly, monthly, and yearly rental models.
 - **Designed a rule-based availability computation engine** combining contract types (fixed, fixed-with-rollover, month-to-month), stay types, and occupancy states to eliminate booking conflicts across thousands of units.
 - Modeled **contract and occupancy state** machines (booked, occupied, under-notice, shifting, moved-out, shifted) to handle real-world edge cases such as early move-outs, tenant shifts, and rollover extensions.
 - Built backend services powering an **availability calendar** with past, present, and future visibility at property, unit, and category levels, enabling operational planning and inventory optimization.
 - **Optimized critical APIs and MongoDB queries** used in booking and availability flows, significantly reducing response latency under concurrent access.
 - Integrated external payment providers (Razorpay, Stripe, Hitpay) using **idempotent APIs and failure-safe work-flows**, supporting 10,000+ users and minimizing payment inconsistencies.
 - Designed backend modules for mall operations (gatepass, work permits, checklists), reducing manual workflows by 40and improving operational traceability.
 - Built event-driven and serverless workflows using **AWS Lambda, S3, and SQS**, ensuring scalability, fault tolerance, and 99.9
 - Collaborated closely with product and operations teams to translate ambiguous business rules into deterministic backend logic used in production.

Technical Expertise

- **Languages:** Java, JavaScript, TypeScript, SQL
- **Backend Development:** Nodejs, ExpressJS, Mongoose
- **Frontend Development:** ReactJS, html5, css3, Bootstrap, Angular
- **Database & Tools:** MongoDB, MySQL, Git
- **Cloud Technologies:** AWS Lambda, AWS S3, SQS, Docker
- **AI & LLM Technologies:** OpenAI (GPT-4),prompt engineering Copilot,
- **Core Competencies:** Object-Oriented Programming (OOPs), Data Structures and Algorithms, System Design
- **Development Practices:** Microservices, RESTful APIs, CI/CD, Agile

Software Projects

- **Full-Stack E-commerce Platform** (2022)
 - Developed scalable e-commerce solution with Redux state management optimizing data fetch by 60%
 - Implemented performance optimization achieving 2-second page load time
 - **Technologies:** ReactJS, NodeJS, MongoDB, Material-UI
 - **GitHub:** [E-commerce Project](#)
- **Enterprise Inventory Management System** (2023)
 - Engineered real-time inventory tracking system managing 1000+ SKUs
 - Implemented role-based access control reducing manual reconciliation by 70%
 - **Technologies:** AngularJS, NodeJS, MongoDB, RESTful APIs

Education

Year	Degree	Institution	GPA
2019-2023	Bachelor of Technology, Computer Science and Engineering	Government College of Engineering and Ceramic Technology, Kolkata	9.33/10